

Mohammad Mahmudul Hasan, PhD

Researcher, Electrical Engineering and Computer Science
& Associate Professor, Electrical and Electronic Engineering



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UPCE · UITS · Portfolio

SUMMARY

- Academic researcher and educator in electrical engineering and information and communication technology, with a PhD from NTNU and more than 17 years of experience across Bangladesh, India, Norway, and the Czech Republic. My work spans research, teaching, supervision, academic leadership, and professional service, with research interests in wireless communication, RF and microwave systems, sensing technologies, signal processing, and machine learning. I bring experience in peer-reviewed research, funded projects, curriculum development, and student mentoring, with emphasis on methodological development, experimental validation, and collaborative academic work.

EDUCATION

- Norwegian University of Science and Technology (NTNU), Norway** (05.2022 – 06.2025)
Doctor of Philosophy, Information and Communication Technology
 - Research focus: Wireless communication, machine learning, and microwave systems
 - Recognized through nomination for Best Doctoral Thesis in Sensors
- KIIT University, India** (07.2008 – 06.2010)
Master of Technology, Communication Systems Engineering
 - Specialization: Wireless communication and communication systems
 - Completed with first position; recipient of Chancellor's and Founder's Gold Medals
- KIIT University, India** (07.2004 – 06.2008)
Bachelor of Technology, Electronics and Telecommunication Engineering
 - Specialization: Electronics, antenna/RF engineering, and wireless communication

WORK EXPERIENCE

- University of Pardubice (UPCE), Czech Republic** (08.2026 – Present)
 - Researcher, Faculty of Electrical Engineering and Computer Science
- University of Information Technology and Sciences (UITS), Bangladesh** (01.2011 – Present)
 - Associate Professor (On Leave), Electrical and Electronic Engineering 07.2025 – Present
 - Assistant Professor, Electrical and Electronic Engineering 01.2011 – 06.2025
- Norwegian University of Science and Technology (NTNU), Norway** (05.2022 – 06.2025)
 - Researcher, Faculty of Engineering
- Brno University of Technology (VUT), Czech Republic** (08.2024 – 01.2025)
 - Researcher, Department of Automation
- Sanyo Engineering & Construction Inc. (SUNTEC), Japan** (04.2018 – 01.2019)
 - Consultant and Technical Trainer
- KIIT University, India** (02.2009 – 12.2010)
 - Assistant Professor, Electronics and Telecommunication Engineering 07.2010 – 12.2010
 - Teaching Assistant, Electronics and Telecommunication Engineering 02.2009 – 06.2010

RESEARCH PROJECTS AND INITIATIVES

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|---|---|-------------|
| 📖 | Communication, Information and Detection Technologies for Control and Signalling Systems (CIDET) , Project Member, University of Pardubice (CZ.02.01.01/00/23_021/0008402) | 2026 – 2027 |
| 📖 | Machine Learning Assisted Signal Processing for Antenna-Sensor Arrays
Project Member, The Research Council of Norway (353890) | 2024 – 2025 |
| 📖 | Air Quality Monitoring Exploiting Massive Multiple-Input Multiple-Output Antennas
Project Member, The Research Council of Norway (324061) | 2022 – 2025 |
| 📖 | University Comprehensive Academic Management-Software Implementation Project
Project Manager, University of Information Technology and Sciences (UITS) | 2020 – 2022 |
| 📖 | Higher Education Quality Enhancement Project (HEQEP)
Member, University Grants Commission, Bangladesh and the World Bank | 2017 – 2018 |

SELECTED PUBLICATIONS

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|---|---|--------|
| 📖 | Inaamullah Khan, Mohammad Mahmudul Hasan , Michael Cheffena, “Transmitter-Assisted Joint Data-Aided Channel Estimation and PAPR Reduction Scheme in Wireless Fading Channels”, Scientific Reports , vol. 16, no. 1, 8015, 2026 | (2026) |
| 📖 | Mohammad Mahmudul Hasan , Onur Alev, Pavel Skrabanek, Gabriela Soukupová, Fatima Hassouna, and Michael Cheffena, “Microwave MIMO E-Nose for Wireless Communication and Selective Detection of VOC Mixtures with Concentration Estimation”, ACS Sensors , vol. 10. no. 9, pp. 6446–6463, 2025 | (2025) |
| 📖 | Inaamullah Khan, Mohammad Mahmudul Hasan , Michael Cheffena, “A Novel Low-Complexity Peak-Power-Assisted Data-Aided Channel Estimation Scheme for MIMO-OFDM Wireless Systems”, IEEE Open Journal of Signal Processing , vol. 6, pp. 992–1003, 2025 | (2025) |
| 📖 | Onur Alev, Mohammad Mahmudul Hasan , Emel Tuğba Ertuğrul, Selçuk Birdoğan, Okan Özdemir, Eda Goldenberg, Michael Cheffena, “Hydrothermally Synthesized Molybdenum disulfide Nanoflakes: Structural, Electrical, and Antenna-based Gas Sensing Characteristics”, Sensors & Actuators: A. Physical , vol. 393, 116756, 2025 | (2025) |
| 📖 | Mohammad Mahmudul Hasan , Onur Alev, Pavel Skrabanek, and Michael Cheffena, “Molecularly Imprinted Polymer-Based Electronic Nose for Ultrasensitive, Selective Detection and Concentration Estimation of VOC Mixtures”, IEEE Sensors Journal , vol. 25, no. 10, 2025 | (2025) |
| 📖 | Mohammad Mahmudul Hasan , Todd Cowen, Onur Alev and Michael Cheffena, “MIMO Microwave Sensor for Selective and Simultaneous Detection of Methanol and Ethanol Gases at Room Temperature”, IEEE Transactions on Instrumentation & Measurement , vol. 74, 9511613, 2025 | (2025) |
| 📖 | Mohammad Mahmudul Hasan , Onur Alev and Michael Cheffena, “Dual-Functional Antenna Sensor for Highly Sensitive and Selective Detection of Isopropanol Gas Using Optimized Molecularly Imprinted Polymers”, ACS Sensors , vol. 10, no. 3, pp. 2147–2161, 2025 | (2025) |
| 📖 | Mohammad Mahmudul Hasan , Onur Alev, Eda Goldenberg and Michael Cheffena, “MoS ₂ /MoO _x Nanoflake-Based Dual-Functional Antenna Sensors for Highly Sensitive and Selective Detection of Volatile Organic Compounds”, ACS Applied Nano Materials , vol. 7, no. 21, pp. 25065–25077, 2024 | (2024) |
| 📖 | Mohammad Mahmudul Hasan , Todd Cowen and Michael Cheffena, “A Novel Molecularly Imprinted Polymer-Based Carbon Nanotube-Coated Microwave Sensor for Selective Detection of Methanol Gas”, IEEE Sensors Letters , vol. 8, no. 5, 6004904, 2024 | (2024) |

- 📄 **Mohammad Mahmudul Hasan**, Onur Alev, Eda Goldenberg and Michael Cheffena, “A Novel Molybdenum Disulfide-Based High-Precision Microwave Sensor for Methanol Gas Detection at Room Temperature,” **IEEE Microwave and Wireless Technology Letters**, vol. 34, no. 6, pp. 691–694, 2024 (2024)
- 📄 **Mohammad Mahmudul Hasan**, Michael Cheffena, “Adaptive Antenna Impedance Matching Using Low-Complexity Shallow Learning Model”, **IEEE Access**, vol. 11, pp. 74101–74111, 2023 (2023)
- 📄 **Mohammad Mahmudul Hasan**, Michael Cheffena, Slobodan Petrovic, “Physical-layer Security Improvement in MIMO OFDM Systems using Multilevel Chaotic Encryption”, **IEEE Access**, vol. 11, pp. 64468–64475, 2023 (2023)
- 📄 Inaamullah Khan, Michael Cheffena, **Mohammad Mahmudul Hasan**, “Data Aided Channel Estimation for MIMO-OFDM Wireless Systems Using Reliable Carriers”, **IEEE Access**, vol. 11, pp. 47836–47847, 2023 (2023)
- 📄 **Mohammad Mahmudul Hasan**, Mohammad Mahadi Hasan Foad, “Modified Gamma Correction Companding for PAPR Reduction in OFDM Systems Considering Solid State Power Amplifier and Wireless Channels”, **Circuits, Systems, and Signal Processing**, vol. 37, no. 10, pp. 4431–4454, 2018 (2018)
- 📄 **Mohammad Mahmudul Hasan**, Mohammad Mahdi Hasan Faisal, “IGCC for PAPR Reduction in OFDM Systems over the Nonlinearity of SSPA and Wireless Fading Channels”, **Circuits, Systems, and Signal Processing**, vol. 35, no. 8, pp. 2855–2880, 2015 (2015)
- 📄 **Mohammad Mahmudul Hasan**, “PAR Reduction in SU/MU-MIMO OFDM Systems using OPF Precoding over the Nonlinearity of SSPA”, **Wireless Personal Communications**, vol. 83, no. 3, pp. 2225–2248, 2015 (2015)
- 📄 **Mohammad Mahmudul Hasan**, “A Novel CVM Precoding Scheme for PAPR Reduction in OFDM Transmissions”, **Wireless Network**, vol. 20, no. 6, pp. 1573–1581, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “A New PAPR Reduction Scheme for OFDM Systems Based on Gamma Correction”, **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1655–1668, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “A New PAPR Reduction Technique in OFDM Systems Using Linear Predictive Coding”, **Wireless Personal Communications**, vol. 75, no. 1, pp. 707–721, 2014 (2014)
- 📄 **Mohammad Mahmudul Hasan**, “VLM Precoded SLM Technique for PAPR Reduction in OFDM Systems”, **Wireless Personal Communications**, vol. 73, no. 3, pp. 791–801, 2013 (2013)
- 📄 **Mohammad Mahmudul Hasan**, “PAPR Reduction in OFDM Systems Based on Autoregressive Filtering”, **Circuits, Systems, and Signal Processing**, vol. 33, no. 5, pp. 1637–1654, 2013 (2013)

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ACADEMIC LEADERSHIP AND PROFESSIONAL SERVICE

- 📄 **Sensors**, MDPI (2026 – Present)
 - 🕒 **Guest Editor**, *Smart Antenna & MIMO Systems for Wireless Communications*
- 📄 **University of Information Technology and Sciences, Bangladesh** (01.2011 – Present)
 - 🕒 **Advisor**, Information and Communication Technology Cell 01.2018 – Present
 - 🕒 **Head of the Department**, Electronics and Communication Engineering 02.2019 – 04.2022
 - 🕒 **Director**, Information and Communication Technology Cell 12.2020 – 04.2022
 - 🕒 **Editorial Board Member**, Journal of Science and Engineering 10.2020 – 04.2022
 - 🕒 **Member**, Internal Quality Assurance Cell (IQAC) 07.2017 – 07.2018
 - 🕒 **Member**, Finance Committee 03.2020 – 04.2022
 - 🕒 **Member**, Academic Council 02.2019 – 04.2022

PROFESSIONAL TRAINING AND CERTIFICATIONS

- 📄 **Brno University of Technology, Czech Republic** (08.2024 – 01.2025)
Professional Competence, Electrical Engineering, NV No. 194/2022 Coll.
- 📄 **GrameenPhone Ltd., Bangladesh** (07.2007 – 10.2007)
Intern Engineer, Transmission Planning Division
- 📄 **All India Radio & Doordarshan (Prasar Bharati), India** (05.2007 – 07.2007)
Industrial Training, Broadcasting Corporation of India
- 📄 **Red Hat Bhubaneswar, India** (04.2006 – 07.2006)
Industrial Training, Red Hat Linux RHEL 4

GRANTS & AWARDS

- 🏆 **Research Recognition** (06.2025)
Nominated for Best Doctoral Thesis in Sensors
Norwegian University of Science and Technology
- 🏆 **Research Grant** (08.2024)
Awarded 146 000 NOK
The Research Council of Norway
- 🏆 **Chancellor's Gold Medal** (12.2010)
Awarded for securing the highest CGPA (10/10)
KIIT University, India
- 🏆 **Founder's Gold Medal** (12.2010)
Awarded for securing the first position in Master of Technology
KIIT University, India
- 🏆 **Chancellor's Merit Scholarship** (2004 – 2010)
Awarded for academic excellence, Bachelor's and Master's studies
KIIT University, India

LANGUAGE PROFICIENCY

- 👍 Bengali (native), English (fluent), Hindi (fluent), Arabic (A1), and Norwegian (A2/B1)

CITIZENSHIP & RESIDENCY

- 📄 Bangladeshi citizen; Norwegian permanent resident since 2022; Czech Republic temporary resident since 2026

REFERENCES

👤 Dr. Michael Cheffena

(PhD Supervisor)

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👤 Dr. Alok Mishra

(Distinguished Professor)

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